Matthew Yu

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UW-Madison graduate (B.S. Computer Science and Data Science) with full-stack experience and an interest in backend engineering, low-level systems programming (compilers and operating systems), and distributed systems through C++.

Education

University of Wisconsin - Madison Madison, WI
B.S. Computer Science, Data Science *May 2026*
GPA: 3.56

Technical Skills

Languages: C++, Java, C#, Python, JavaScript, TypeScript, SQL, Bash, HTML, CSS

Backend & Data: ASP.NET Core, Django, REST APIs, PostgreSQL, SQL Server, SQLite

Frontend: React, Next.js

Infrastructure & Tools: Docker, Linux, Git, GitHub Actions, Claude Code

Experience

WEM Automation - A Valstone Company Pewaukee, WI
Software Engineer Intern *Jan 2026 – June 2026*

- Architected a full-stack manufacturing analytics platform (React, TypeScript, ASP.NET Core, SQL Server), giving plant managers real-time KPI visibility across 200+ daily production batches.
- Designed backend data architecture, REST APIs, and a dedicated SQL aggregation layer that turn raw operational data into query-efficient time-series KPIs for responsive dashboards.
- Analyzed post-installation equipment data from a feed mill and built a C# app that turns a selected data file into a PDF report with KPI visualizations, cutting analysis time from 2 hours to minutes and surfacing inefficiencies for plant stakeholders.

Green Leaf Cuisine – Restaurant Website & Deployment Pipeline *Summer 2026*
greenleafcuisine.com

- Built and deployed a responsive React/TypeScript restaurant website from scratch, developing reusable components for 100+ menu items, image galleries, accordions, and mobile navigation across multiple pages.
- Designed and implemented a GitHub Actions CI/CD pipeline with automated pull-request build validation and deployments to staging and production environments.

Selected Projects

CHIP-8 Emulator *C++, SDL2, Make*
github.com/MatthewYu402/CHIP-8 *2026*

- Built a CHIP-8 emulator from scratch in C++20, implementing all 35 opcodes in a fetch-decode-execute loop over 4 KB of memory and 16 8-bit registers, with SDL2 handling the 64×32 display, 16-key keypad input, and audio.
- Decoupled CPU speed from the 60 Hz delay and sound timers with a fixed-rate frame loop, and implemented blocking key input (FX0A) as a press-and-release state machine.
- Passes the Timendus CHIP-8 test suite (opcodes, flags, keypad, and sound) and runs classic games such as Pong; configurable platform quirks (COSMAC VIP vs. CHIP-48) are in progress.

LeetCode Visualizer – Language-Agnostic Code Tracing Tool *C++, Python, TypeScript, React*
github.com/MatthewYu402/LeetCode-Visualizer *In progress*

- Designing a visualizer that takes an unmodified LeetCode `Solution` class, lets users step through an example line by line, and draws arrays, hash maps, linked lists, trees, and graphs as they change, to build intuition and debug solutions.
- Specified a language-neutral JSON trace format with a JSON Schema as the only contract between backend and frontend, so each language needs one adapter and each renderer works for all languages ($L \times R$ integrations reduced to $L + R$).
- Planning C++ support first via Clang-based source instrumentation, then Python and others via runtime hooks or the Debug Adapter Protocol, running untrusted code in a sandbox with CPU, memory, and time limits and validating each trace outside it.

Senior Capstone Project – Neurodiversity Therapeutics Tablet Application *Python, Kivy, SQLite*
Rosenberg Lab

- Built a Python/Kivy application with Pupil Labs eye-tracking integration to identify autism subgroups based on contextual learning vs. visual saliency patterns, enabling personalized therapeutics.
- Implemented real-time behavioral data collection on SQLite with a multi-threaded architecture, synchronizing response times, accuracy, and gaze data across 250+ trials.
- Collaborated with UW-Madison neuroscience researchers on a 5-person cross-functional Agile/Scrum team using Jira, Figma, Slack, and Git.

Activities

WSUM 91.7 FM – Radio Show Host *May 2024 – May 2026*

- Hosted a weekly live radio program for a campus-wide audience at WSUM, the Top College Radio Station (2026).